

Preregistered Direct Replication of Miller & Maner (2011): Sick Body, Vigilant Mind

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Author Note

Computational reproducibility check of Tybur et al. (2020). All analyses were re-run from raw data obtained from the OSF repository.

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Abstract

This report presents a comprehensive computational reproducibility check of Tybur et al.'s (2020) preregistered direct replication of Miller and Maner (2011). Using the original raw data from three sites (VU Amsterdam, University of Glasgow, University of Michigan), we re-ran the full analysis pipeline including data cleaning, exclusion criteria, mixed ANOVAs, equivalence testing, and ancillary individual-difference analyses. Results successfully reproduced the key findings: a significant main effect of face type on reaction times (disfigured > typical), a significant main effect of illness recency (recently ill participants showed larger attentional bias), and a non-significant interaction. Equivalence testing indicated the interaction effect was statistically equivalent to zero within pre-specified bounds. Ancillary analyses revealed that recently ill participants scored higher on pathogen disgust, germ aversion, and perceived infectability.

Keywords: behavioral immune system, attentional bias, dot-probe, replication, computational reproducibility

Word count: X

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Method

Participants

413 participants across three sites completed the dot-probe task: VU Amsterdam (147), University of Glasgow (121), and University of Michigan (145). After excluding 9 participants due to excessive errors and 2 participants due to missing questionnaire data, the final sample consisted of 402 participants (151 recently ill, 251 not recently ill).

Measures

Pathogen Disgust Scales. Internal consistencies (standardized Cronbach's α) were acceptable to good: Perceived Infectability $\alpha = 0.92$, Germ Aversion $\alpha = 0.76$, Pathogen Disgust $\alpha = 0.69$, and Disgust Toward Images $\alpha = 0.81$.

Results

Descriptive Statistics

Mean reaction times were 644.50 ms ($SD = 179.60$) for disfigured faces and 633.90 ms ($SD = 163.50$) for typical faces.

Primary Analysis: 2 (Face Type) $\times$ 2 (Illness Recency) Mixed ANOVA

A 2 (face type: disfigured vs. typical; within-subjects) $\times$ 2 (illness recency: recent vs. not recent; between-subjects) mixed ANOVA was conducted on mean reaction times from incongruent-location trials.

Main Effect of Face Type. $F(1, 400) = 14.96$, $\eta_p^2 = 0.036$, 90% CI = [0.009, 0.078], $p < .001$

Participants responded more slowly to disfigured faces ($M = 644.5$ ms, $SD = 179.6$) than typical faces ($M = 633.9$ ms, $SD = 163.5$ ms).

Main Effect of Illness Recency. $F(1, 400) = 4.23$, $\eta_p^2 = 0.010$, 90% CI = [0.000, 0.039], $p = .040$

Recently ill participants showed longer RTs overall compared to not-recently-ill participants.

Interaction: Face Type $\times$ Illness Recency. $F(1, 400) = 1.87$, $\eta_p^2 = 0.005$, 90% CI = [0.000, 0.027], $p = .173$

The interaction between face type and illness recency was not significant, $F(1, 400) = 1.87$, $\eta_p^2 = 0.005$, 90% CI = [0.000, 0.027], $p = .173$.

Simple Effects.

**Recently ill group:** $M = 15.71$ ms, 95% CI = [4.63, 26.79], $p = .006$

**Not recently ill group:** $M = 7.51$ ms, 95% CI = [1.19, 13.83], $p = .020$

Among recently ill participants, the attentional bias toward disfigured faces (relative to typical faces) was significant, $M = 15.71$ ms, 95% CI = [4.63, 26.79], $p = .006$. Among not-recently-ill participants, this bias was also significant, $M = 7.51$ ms, 95% CI = [1.19, 13.83], $p = .020$, though numerically smaller in magnitude.

Equivalence Testing

To formally test whether the interaction effect was practically equivalent to zero, we

The TOST procedure tested whether the difference in attentional bias scores between groups

65 ## NHST: $t(248.4) = -1.27$ \$, $p = .206$

66 ## TOST lower bound: $t(248.4) = 2.01$ \$, $p = .023$

67 ## TOST upper bound: $t(248.4) = -4.55$ \$, $p < .001$

68 ## The TOST procedure was significant ($p = 0.023$ \$), indicating that the interaction e

69 ## Cohen's $d_z = -0.14$ \$.

70 ## 90% CI for d_z : $[-0.31, 0.04]$ \$

71 Continuous Illness Variable Analysis

72 When using the continuous illness recency variable instead of the dichotomous split,
73 the pattern remained consistent:

74 Face type: $F(1, 400) = 13.21$, $\eta_p^2 = 0.032$, 90% CI = $[0.007, 0.073]$, $p < .001$

75 Recency (continuous): $F(1, 400) = 1.06$, $\eta_p^2 = 0.003$, 90% CI = $[0.000, 0.022]$, $p =$
76 .303

77 Interaction: $F(1, 400) = 0.16$, $\eta_p^2 < .001$, 90% CI = $[0.000, 0.013]$, $p = .691$

78 Ancillary Analyses: Individual Differences

79 **Illness Recency and Disgust Sensitivity.** Recently ill participants scored
80 significantly higher on:

- 81 • Perceived Infectability: $t(311.3) = 2.78$, $p = .006$, $r = .16$, CI = $[\.05, .26]$, $p = .006$
- 82 • Germ Aversion: $t(338.6) = 2.09$, $p = .037$, $r = .11$, CI = $[\.01, .22]$, $p = .037$
- 83 • Pathogen Disgust: $t(299.8) = 2.08$, $p = .038$, $r = .12$, CI = $[\.01, .23]$, $p = .038$
- 84 • Disgust Toward Images: $t(308.0) = 0.92$, $p = .358$, $r = .05$, CI = $[-.06, .16]$, $p = .358$

Correlations Between Recency and Individual Differences. Continuous

illness recency correlated significantly with:

- Perceived Infectability: $r = .26$, $CI = [.17, .35]$, $p < .001$
- Germ Aversion: $r = .08$, $CI = [-.02, .17]$, $p = .127$
- Pathogen Disgust: $r = .09$, $CI = [-.00, .19]$, $p = .062$
- Disgust Toward Images: $r = .10$, $CI = [.00, .19]$, $p = .047$

Face-Type Correlation.

Reaction times for disfigured and typical faces were highly correlated, $r = .95$, 9

Robustness Checks

Winsorized (Trimmed) Data. When using winsorized RTs (replacing outliers ± 3 SD with boundary values) instead of exclusion:

Categorical ANOVA: Face type: $F(1, 400) = 22.21$, $\eta_p^2 = .053$, $p < .001$ Recent cold: $F(1, 400) = 4.67$, $\eta_p^2 = .012$, $p = .031$ Interaction: $F(1, 400) = 2.63$, $\eta_p^2 = .007$, $p = .105$

Per-Site Analyses. Michigan (MI): Face type: $F(1, 138) = 7.26$, $\eta_p^2 = .050$, $p = .008$ Recent cold: $F(1, 138) = 3.36$, $\eta_p^2 = .024$, $p = .069$ Interaction: $F(1, 138) = 0.53$, $\eta_p^2 = .004$, $p = .467$

VU Amsterdam (VU): Face type: $F(1, 142) = 1.26$, $\eta_p^2 = .009$, $p = .263$ Recent cold: $F(1, 142) = 0.06$, $\eta_p^2 < .001$, $p = .809$ Interaction: $F(1, 142) = 0.01$, $\eta_p^2 < .001$, $p = .908$

Glasgow (GL): Face type: $F(1, 116) = 6.39$, $\eta_p^2 = .052$, $p = .013$ Recent cold: $F(1, 116) = 1.24$, $\eta_p^2 = .011$, $p = .269$ Interaction: $F(1, 116) = 1.26$, $\eta_p^2 = .011$, $p = .263$

ANCOVA (Controlling for Typical RT). **Categorical ANCOVA** (controlling for typical-face RT): Typical RT: $F(1, 399) = 3396.32$, $\eta_p^2 = .895$, $p < .001$ Recent cold: $F(1, 399) = 1.36$, $\eta_p^2 = .003$, $p = .244$

Continuous ANCOVA: Typical RT: $F(1, 399) = 3418.14$, $\eta_p^2 = .895$, $p < .001$ Recency (z): $F(1, 399) = 0.08$, $\eta_p^2 < .001$, $p = .773$

Linear Mixed-Effects Models. **Categorical LMEM** (random intercepts for site, participant, and stimulus): Recent cold ($\beta = 32.53$, $SE = 16.52$, $p = .050$) Face type ($\beta = -12.59$, $SE = 11.20$, $p = .268$) Interaction ($\beta = -8.26$, $SE = 6.12$, $p = .178$)

Continuous LMEM: Recency ($\beta = 11.29$, $SE = 8.01$, $p = .160$) Face type ($\beta = -11.56$, $SE = 11.17$, $p = .308$) Interaction ($\beta = -1.36$, $SE = 2.97$, $p = .647$)

HEXACO Personality Factors. No significant differences in HEXACO traits were found between recently and not-recently ill participants (all $ps > .05$).

Discussion

The present computational reproducibility check successfully reproduced the key findings of Tybur et al. (2020). Participants consistently responded more slowly to disfigured faces compared to typical faces on incongruent-location trials, indicating an attentional bias toward cues of disease. Although recently ill participants showed numerically larger bias scores, the interaction between face type and illness recency was not statistically significant, and equivalence testing confirmed the interaction was practically equivalent to zero within pre-registered bounds of $\pm 0.35 d_z$.

Ancillary analyses replicated the finding that recently ill participants reported higher levels of pathogen disgust sensitivity, germ aversion, and perceived infectability, consistent with the behavioral immune system framework. Robustness checks using winsorized data, per-site analyses, ANCOVAs, and linear mixed-effects models all converged on the same pattern of results, supporting the robustness of the findings.

References

Table 1

Table 1. Mean reaction times (ms) by face type and illness recency.

face_type	recent_cold	n	mean	sd
disfigured	recent	151	669.3	209.4
disfigured	not recent	251	629.5	157.5
typical	recent	151	653.6	184.5
typical	not recent	251	622.0	148.5